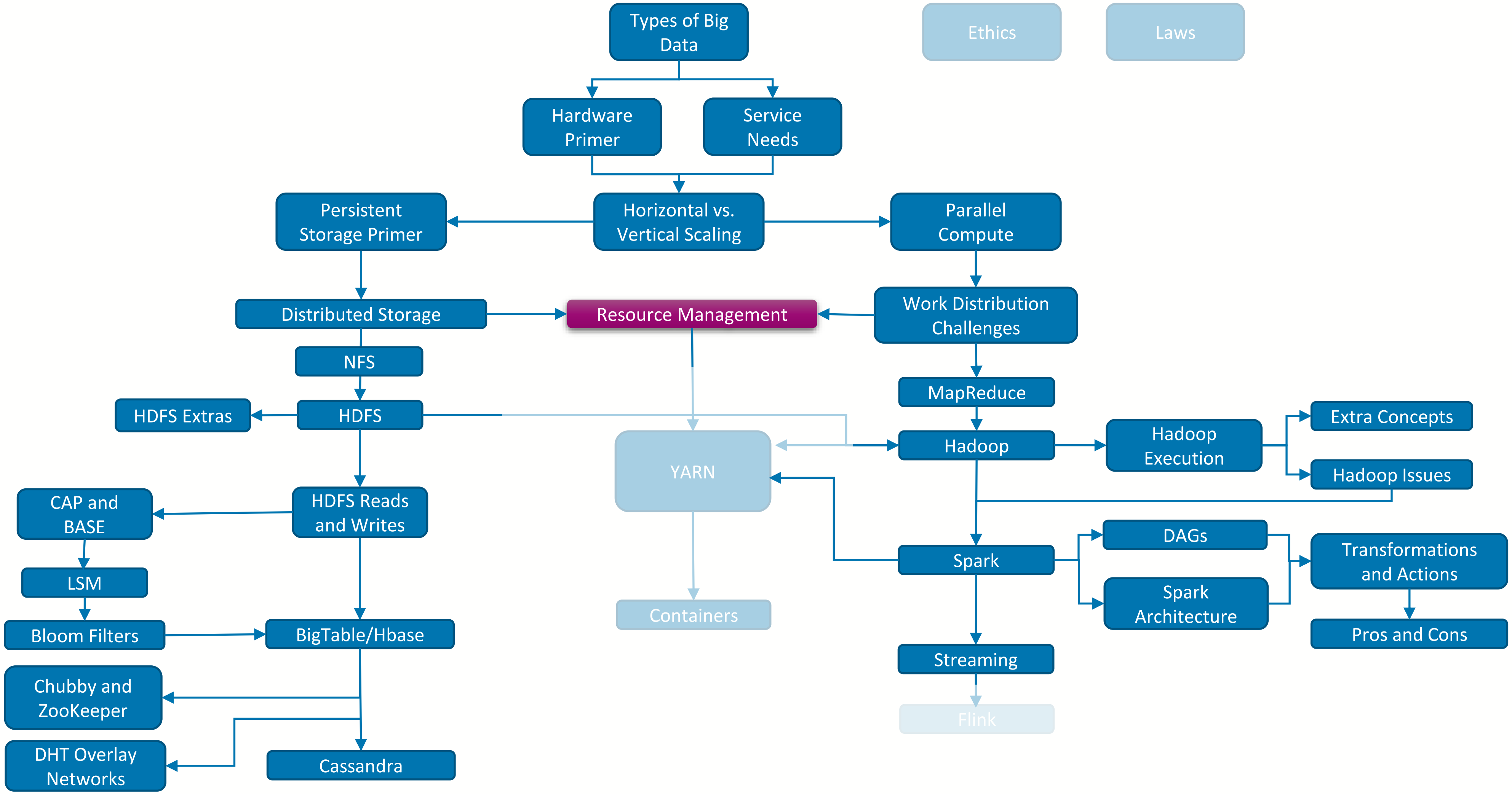
University of Glasgow

W9-03: Cluster Management Challenges

COMPSCI4064 (H) and COMPSCI5088 (M)

Dr. Richard McCreadie

Where are we?



Types of Big Data
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Cluster Management

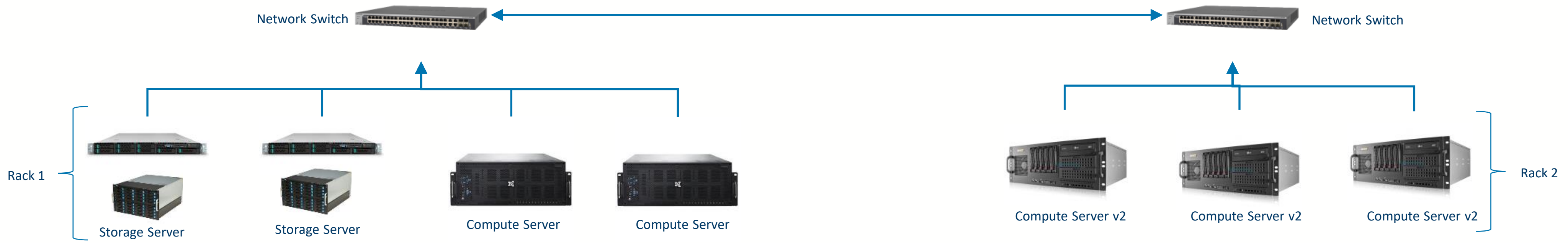
- So far in this course we have presented Big Data platforms at a service level
  - Each platform had services that would be deployed on physical hardware, e.g. the Spark Master, or HDFS Name Node

- However, we have somewhat glossed over a practical challenge…
  - How do we get those services deployed in the first place!

An Example Cluster

- Lets imagine a physical compute cluster
  - Since we are working in Big Data and want horizontal scaling capability, we have a cluster of commodity machines
  - We bought Rack 1 initially, with two file servers and two compute servers all connected on a single switch
  - A year later we bought Rack 2, containing three next generation compute servers that are twice as fast

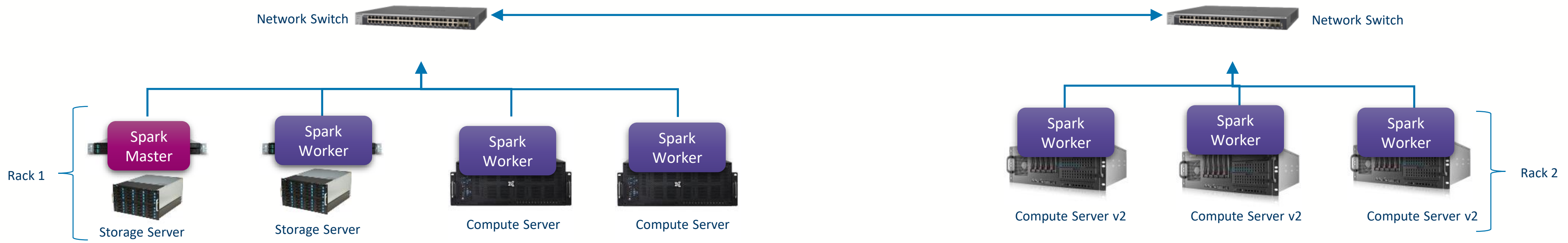
- If I wanted to deploy a Big Data platform on this, e.g. Spark, how would I do it?



Option 1: Do it Manually!

- The most obvious option would be to manually install the services yourself
  - Log on to the machine
  - Install the dependencies
  - Start the service from the command line

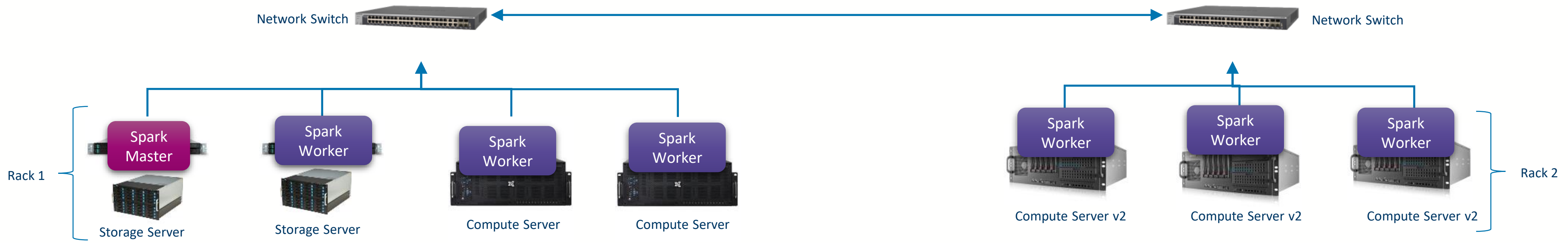
Anyone getting PTSD from getting Spark to work on a single machine?



Problems

- This seems like a lot of work… and the larger the cluster the more work it is going to be
- Also I am going to need to set up all the networking
  - Configurating Spark workers so they can find the master
- What about my data… its all stored in Rack 1
  - Is network latency to Rack 2 an issue?
- Will Spark distinguish between the v1 and v2 compute servers?
  - (Sort of… it will detect core count differences, but not CPU speed differences)

- Most importantly… **what if Spark is not the only thing running here?**



Shared Clusters

- Buying compute clusters are expensive, and the work needing done in a typical company is diverse
  - This means that practically many people will want to use a cluster at one time

- Primary Challenges:
  - How do we avoid driver/library versioning and incompatibility issues amongst the software that is running?
    - Sandbox your software (e.g. via the JVM, Virtual Machine or Container)
  - How do we share/distribute resources amongst the users?
    - Use a Resource Management System (e.g. YARN or Kubernetes)

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Course Coordinator
Dr. Richard McCreadie

Email: richard.mccreadie@glasgow.ac.uk
Room: SAWB 304

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